PCB Design Using KiCad

From schematic to fabrication — a hands-on engineering workflow

Visiting Faculty Lecture • Engineering College Programme

Audience: Beginner → Intermediate | Industry-oriented

Open-source EDA • Schematic + Layout + 3D + Gerber • Manufacturing-ready

Introduction to PCB Design

What a PCB is, why it matters, where it shows up

PCBs — From Concept to Critical Component

What	Why	Types	Where
Printed Circuit Board Mechanical substrate + conductive copper paths that connect components.	Replaces hand-wired circuits. Reliable, mass-producible, compact, repeatable.	• Single-sided • Double-sided • Multilayer (4–32) • Flex / Rigid-flex • HDI / Microvia	Phones, laptops, EVs, IoT, medical, aerospace, robotics, automotive, industrial automation.

What is KiCad?

The open-source EDA suite that runs production hardware

Open-source Electronic Design Automation

- Free, libre, and cross-platform (Windows / macOS / Linux)
- Maintained by CERN, the KiCad Services Corp., and a global community
- Production-grade — used by startups, labs, and Fortune-500 hardware teams
- No node/seat limits, unlimited layers, unlimited pin counts
- Native Python scripting & a plugin ecosystem
- Full toolchain: schematic, PCB, 3D, simulation, fabrication output

Quick Facts

First release	1992
Latest stable	KiCad 8.x / 9.x
License	GPL v3
Languages	20+ localizations
File format	Plain text (Git-friendly)
Cost	₹0 — forever

Features of KiCad

Everything you need under one roof

KiCad — Core Features

Eeschema
Schematic capture with hierarchical sheets

Symbol Editor
Build & customize component symbols

PcbNew
Powerful PCB layout & interactive router

Footprint Editor
Create & manage landpatterns

3D Viewer
Realistic 3D render for mechanical fit

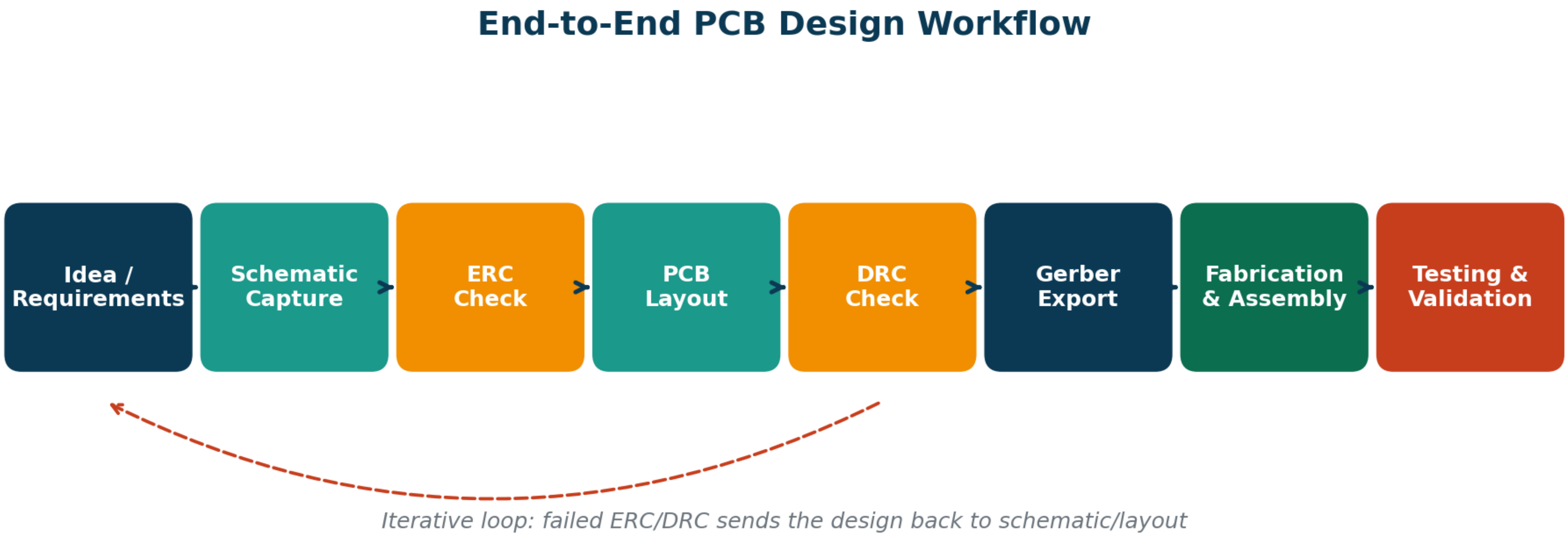
Gerber Viewer
Inspect manufacturing outputs

SPICE Sim
Built-in ngspice simulation

Python API
Scripting, plugins, automation

PCB Design Workflow

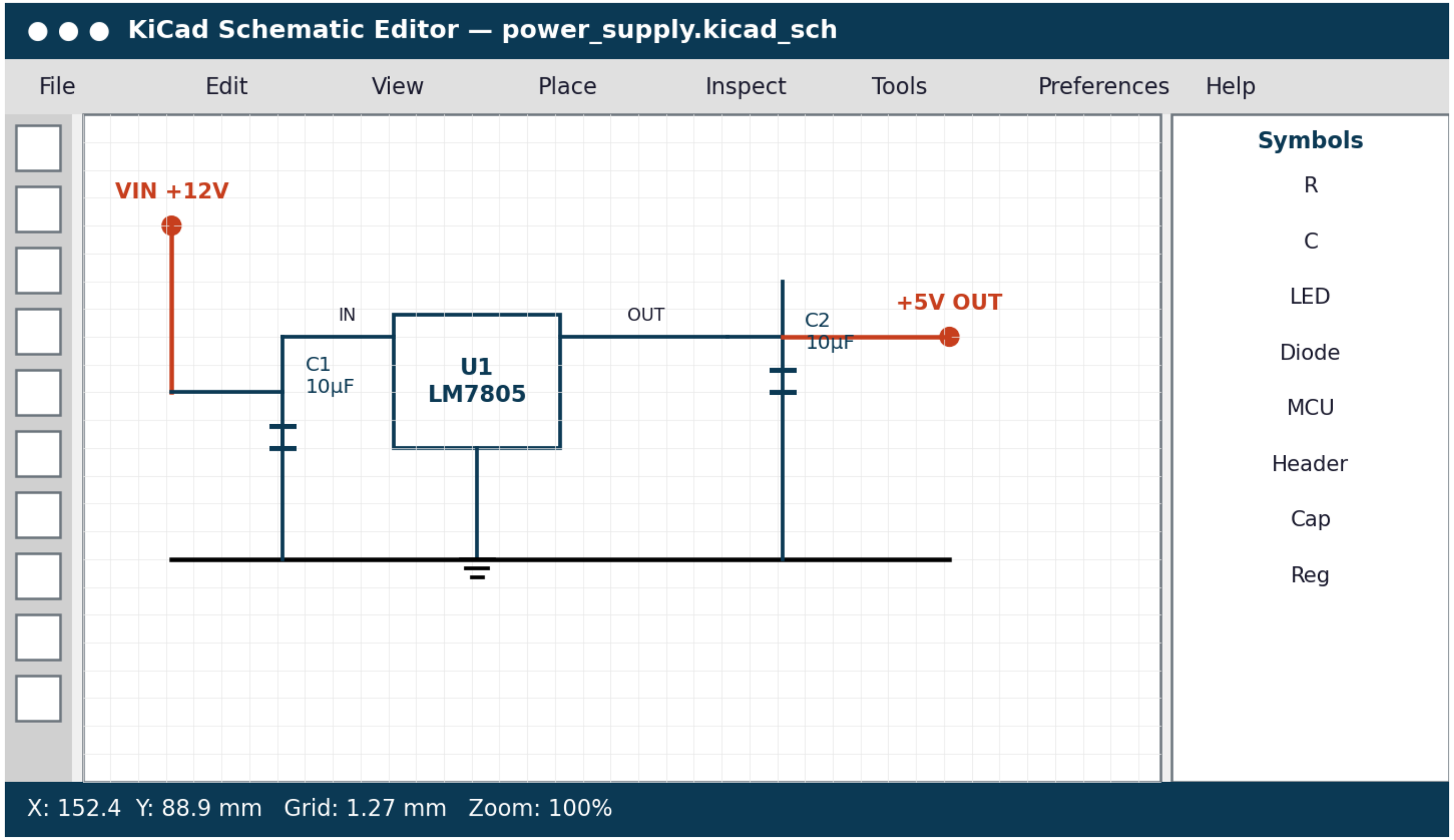
The eight-step flow every engineer follows



Each ERC/DRC failure loops you back — design is iterative, not linear.

Electronic Schematic Design

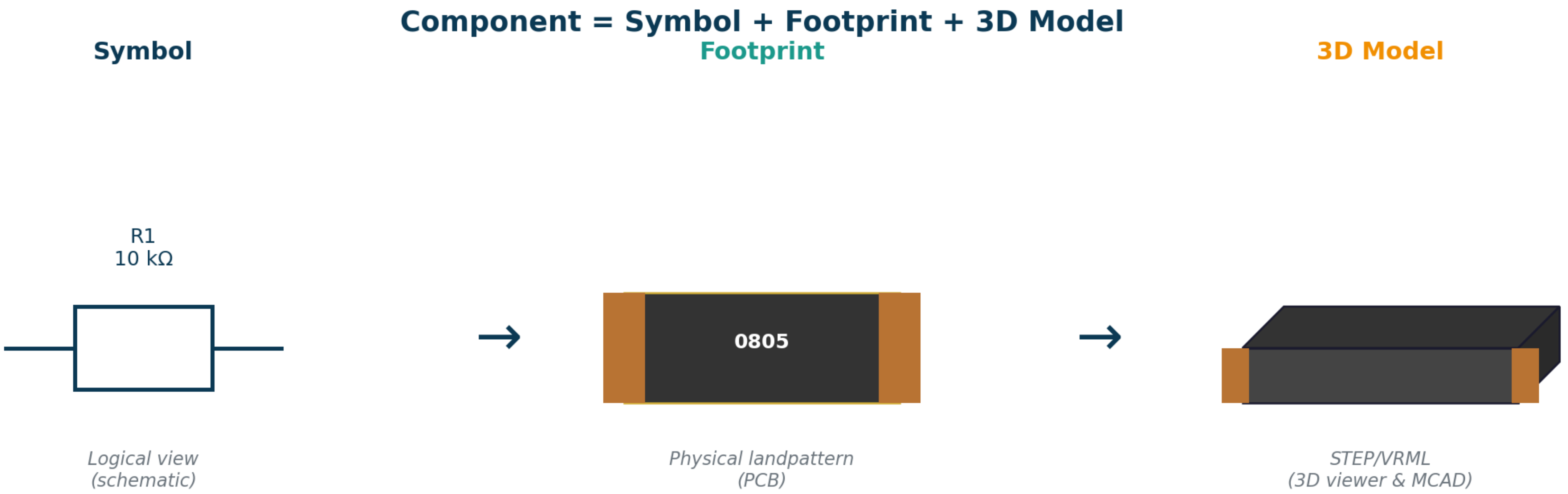
Eeschema — capturing intent before geometry



- Symbols represent function — not physical shape
- Wires define electrical connectivity (nets)
- Power flags mark VCC / GND for ERC
- Hierarchical sheets for large designs
- Annotation: R1, C1, U1 … auto-assigned
- BOM export per ref-des & value
- Cross-probes to PCB editor

Component Libraries & Footprints

Symbol + Footprint + 3D model = a usable part



Where do parts come from?

• KiCad built-in library • SnapEDA / Ultra Librarian / Component Search Engine • Manufacturer-supplied symbols • Your own custom library (always verify against the datasheet!)

ERC and DRC Concepts

Automated checks that catch issues before fabrication

Automated Design Checks

ERC — Electrical Rules Check (Schematic)

- ✓ Unconnected pins
- ✓ Conflicting drivers
- ✓ Missing power flags
- ✓ Output → Output
- ✓ Unused power inputs
- ✓ Hierarchical net mismatches

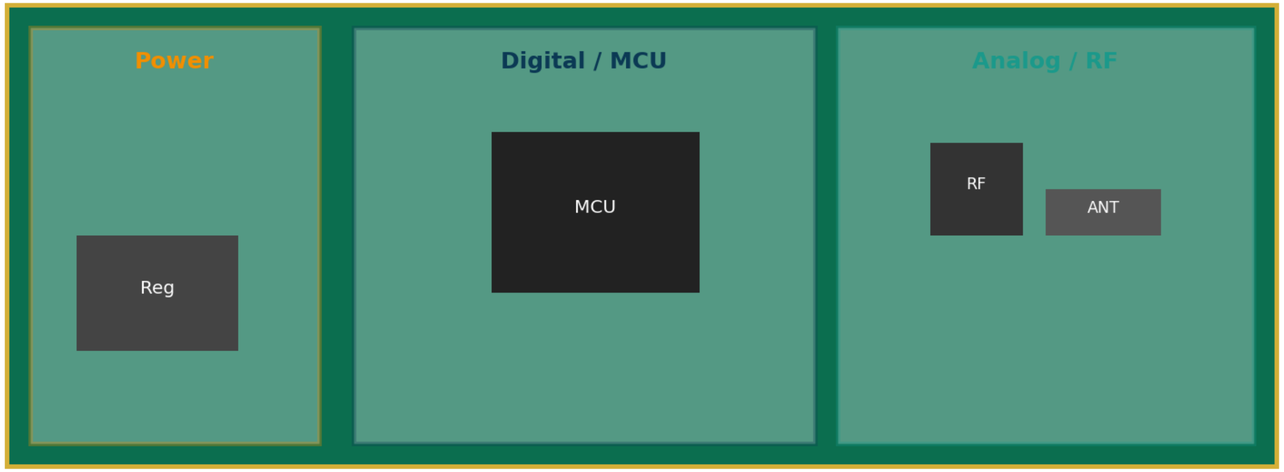
DRC — Design Rules Check (PCB)

- ✓ Trace width violation
- ✓ Clearance violation
- ✓ Hole-to-hole spacing
- ✓ Silk-on-pad
- ✓ Unrouted nets
- ✓ Courtyard overlaps

PCB Layout Fundamentals

Placement before routing — 80% of the work, 20% of the time

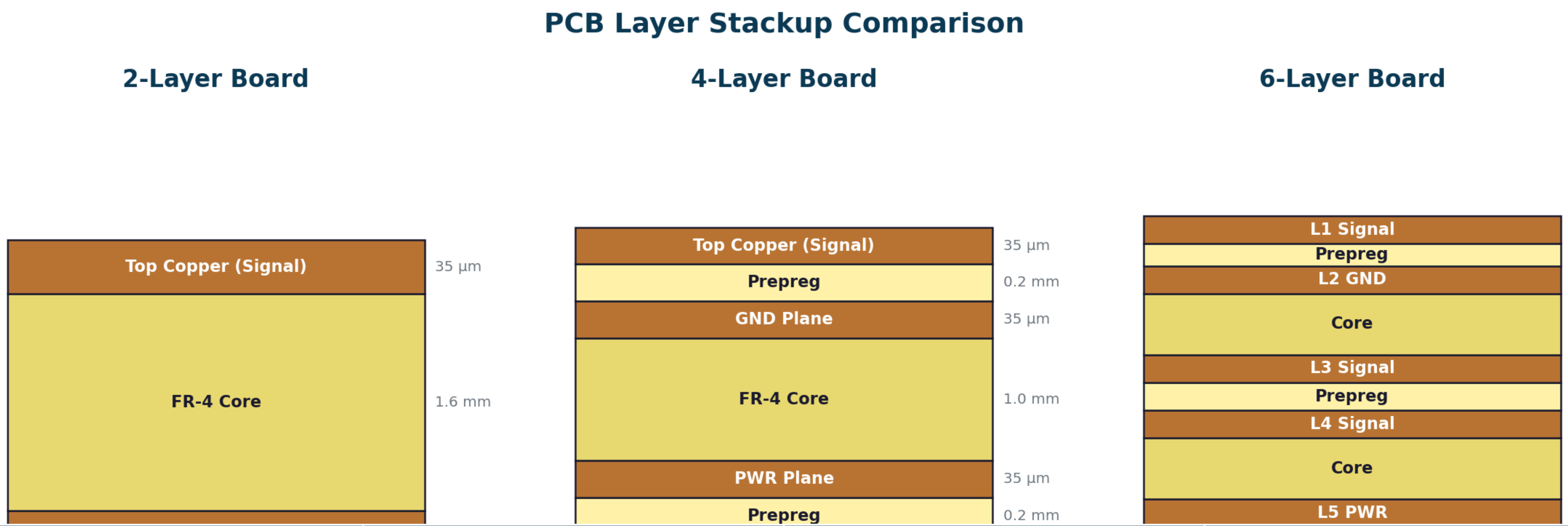
Component Placement Fundamentals



1. Group components by function • 2. Place connectors at board edges • 3. Keep high-speed traces short • 4. Place decoupling caps near IC pins • 5. Separate analog & digital sections • 6. Allow space for thermal & mounting

Layer Stackup

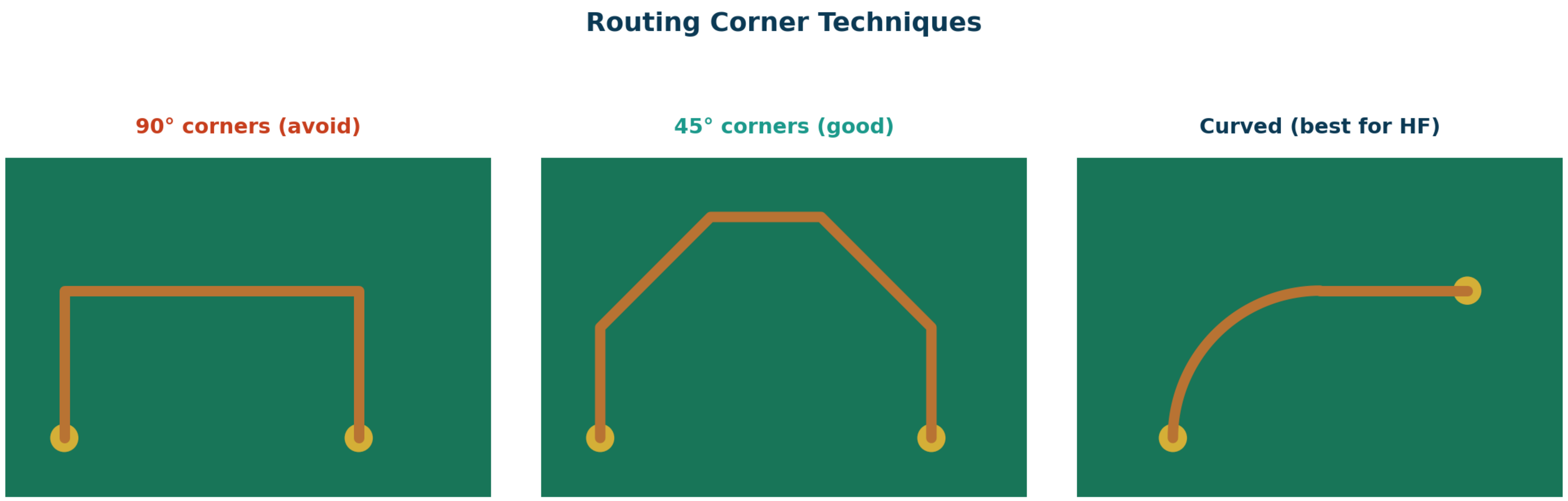
Choosing 2-layer vs 4-layer vs higher — a cost/performance trade



	2-layer	4-layer	6-layer+
Cost (proto, 10 pcs)	₹500–1500	₹2000–5000	₹8000+
Signal integrity	Limited	Good	Excellent
EMI performance	Poor	Good	Best
Best for	Hobby, low-speed	Most products	High-speed, RF, dense

Routing Techniques

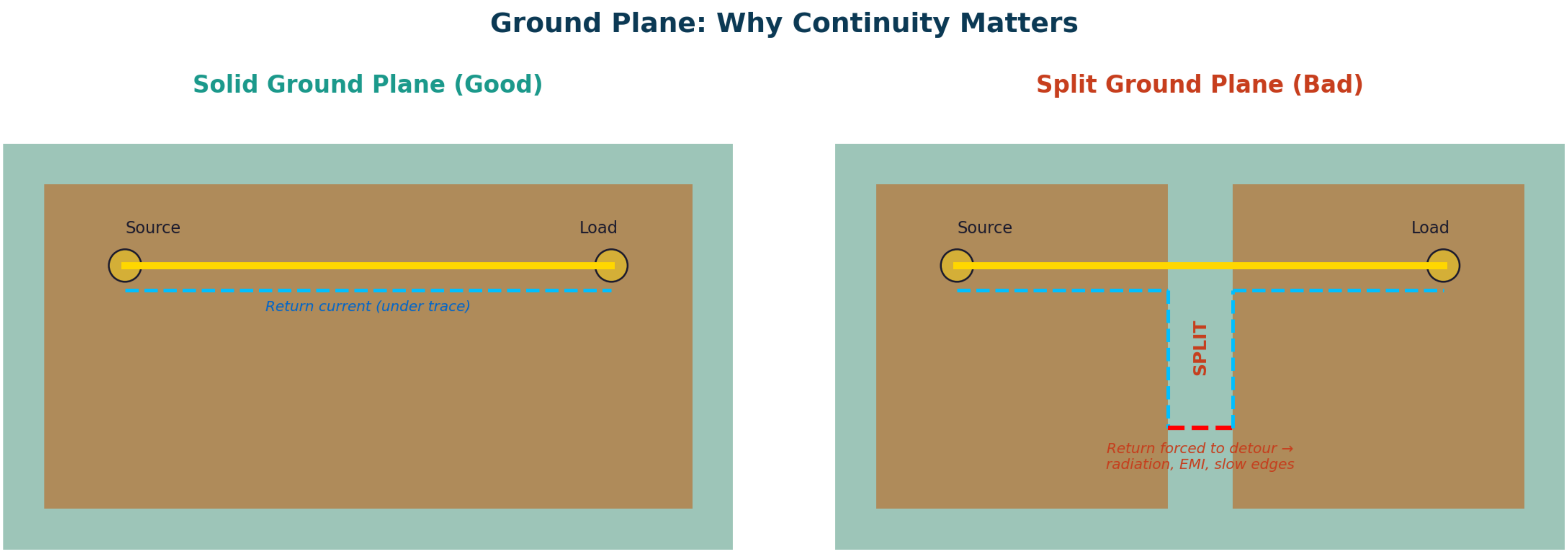
How traces are placed determines signal quality



- Use 45° corners — 90° causes acid traps and impedance discontinuities
- Differential pairs: route as a pair, length-matched within 5 mil (0.13 mm)
- Keep high-speed traces on the layer adjacent to a continuous ground plane
- Power traces sized for current — 10 mil ≈ 1 A on 1-oz copper at 10 °C rise

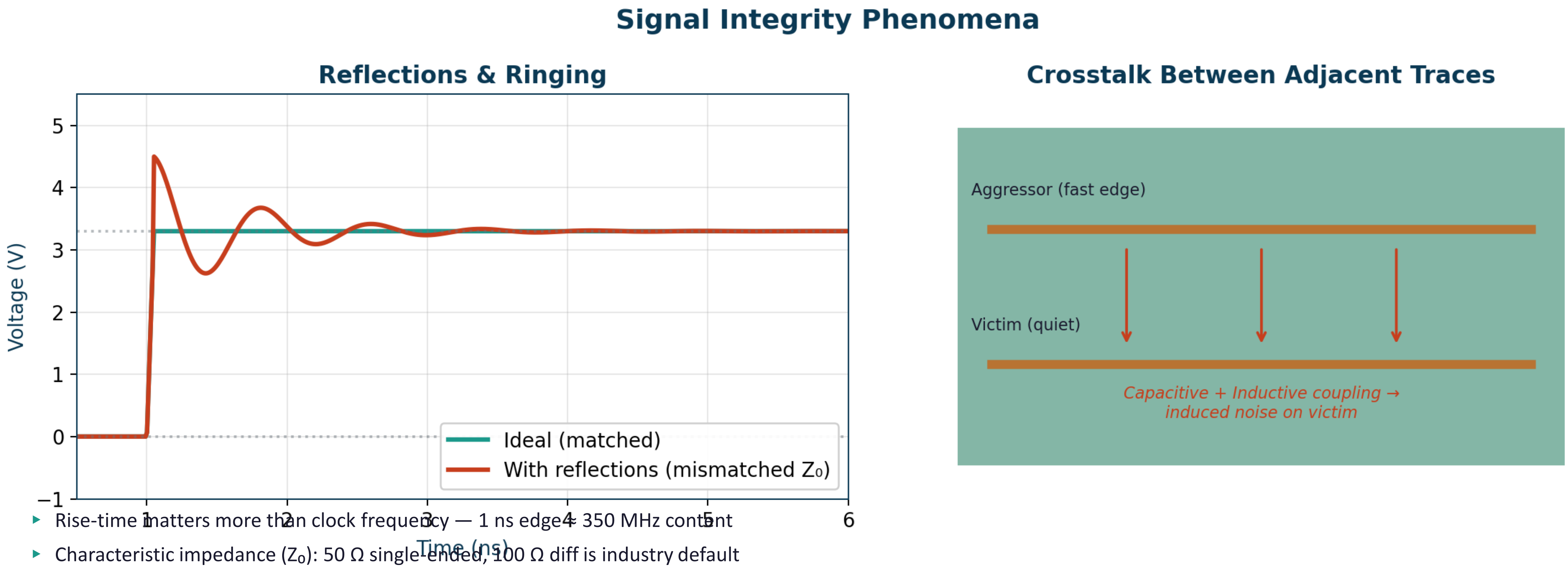
Ground Plane & Power Distribution

Return paths matter as much as forward paths



Signal Integrity Basics

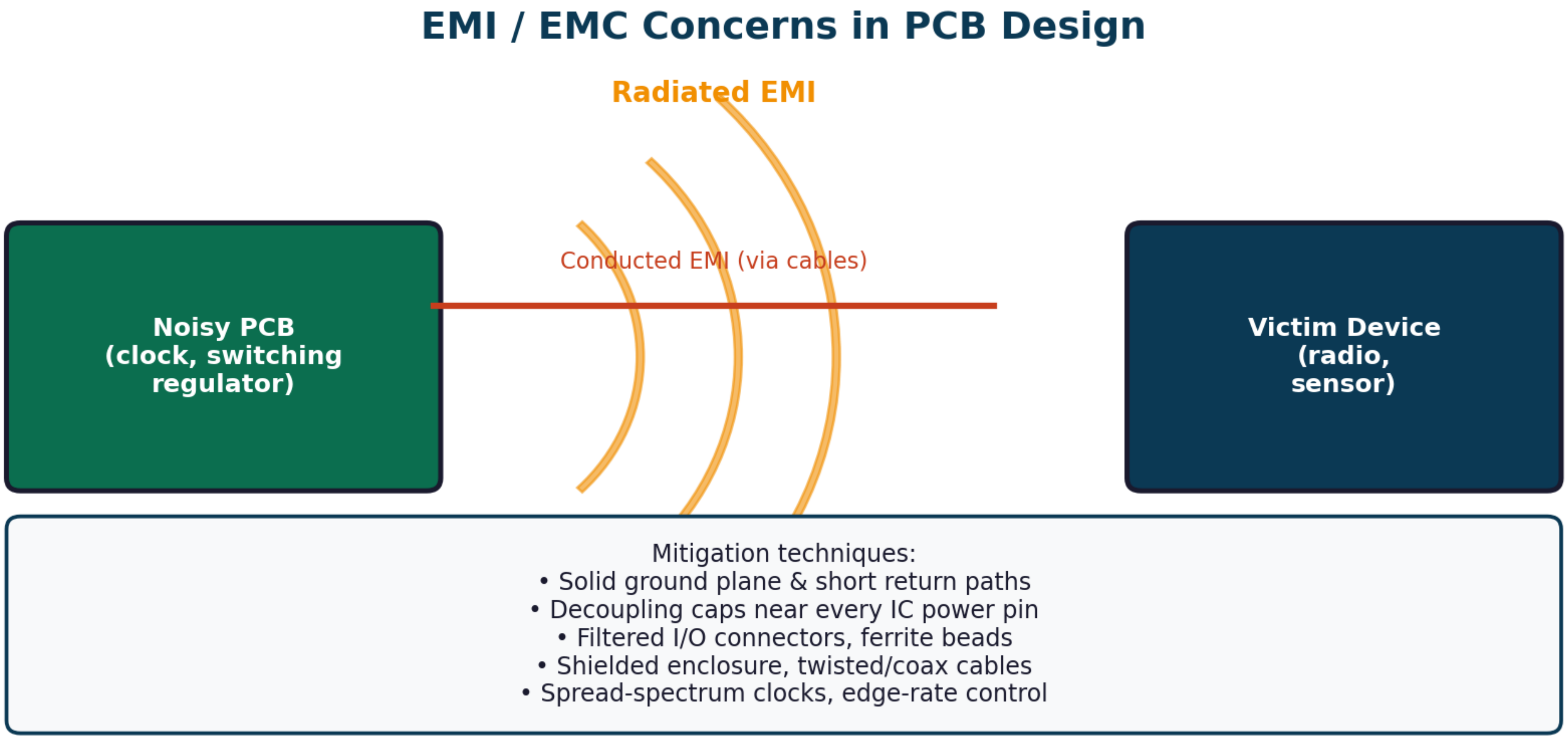
When wires stop behaving like wires



- Rise-time matters more than clock frequency — 1 ns edge ≈ 350 MHz content
- Characteristic impedance (Z_0): 50 Ω single-ended, 100 Ω diff is industry default
- Termination tames reflections — series, parallel, or AC depending on topology
- Crosstalk falls quickly with spacing — keep ≥ 3× trace width between victim & aggressor

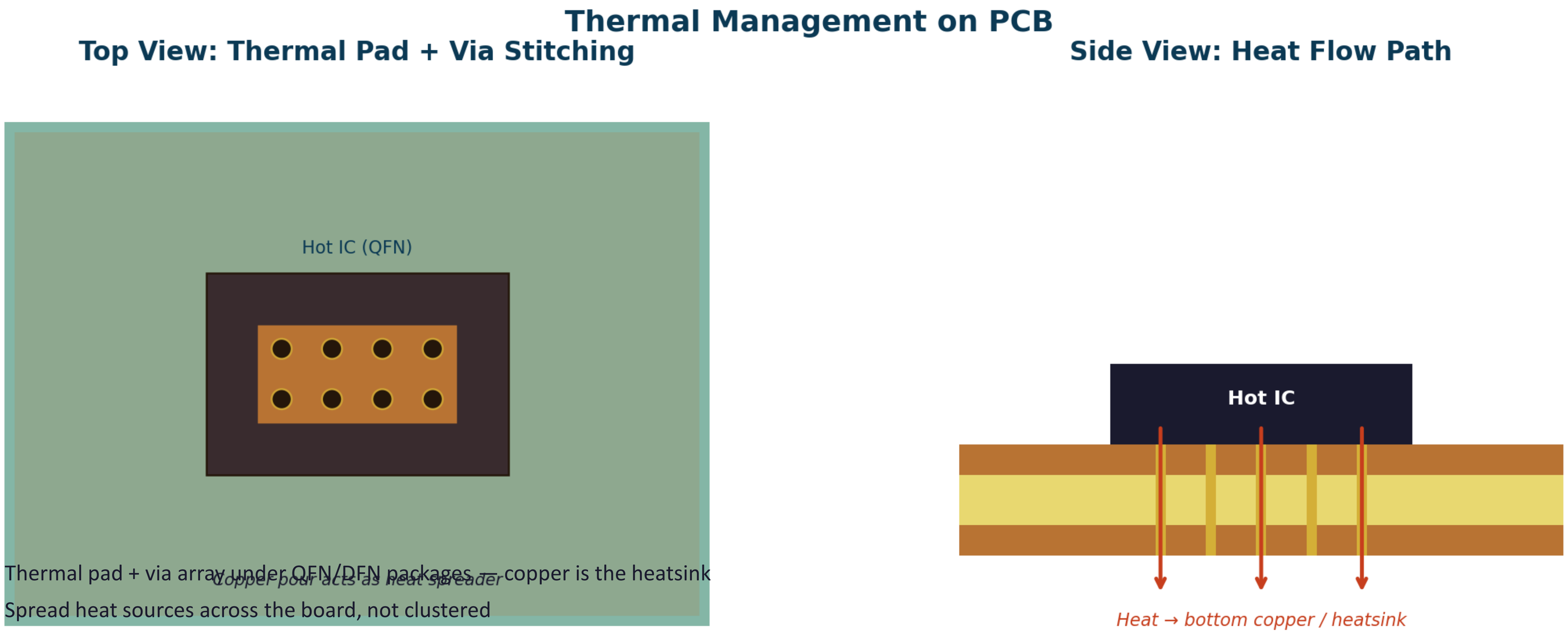
EMI / EMC Considerations

Pass the chamber on the first try



Thermal Management

Heat is the enemy of reliability



- Thermal pad + via array under QFN/DFN packages — copper is the heatsink
- Spread heat sources across the board, not clustered
- Each 10 °C rise halves component lifetime (Arrhenius)
- Calculate: P × θJA = ΔT → size copper area accordingly

Design for Manufacturability (DFM)

Designing within your fab's capabilities

Design for Manufacturability — Key Constraints

Constraint	Note
Trace width ≥ 0.15 mm	Standard fabs ≥ 6 mil
Annular ring ≥ 0.15 mm	Pad − hole radius
Min via drill 0.3 mm	Smaller = HDI cost
Min clearance 0.15 mm	Trace-to-trace/pad
Silkscreen ≥ 0.8 mm text	Readable after print
Solder mask 0.1 mm gap	Prevents bridging
Edge clearance 0.3 mm	Routing/V-cut safety
Standard panel sizes	Cheaper fabrication

Gerber File Generation

The universal handoff format to fabrication

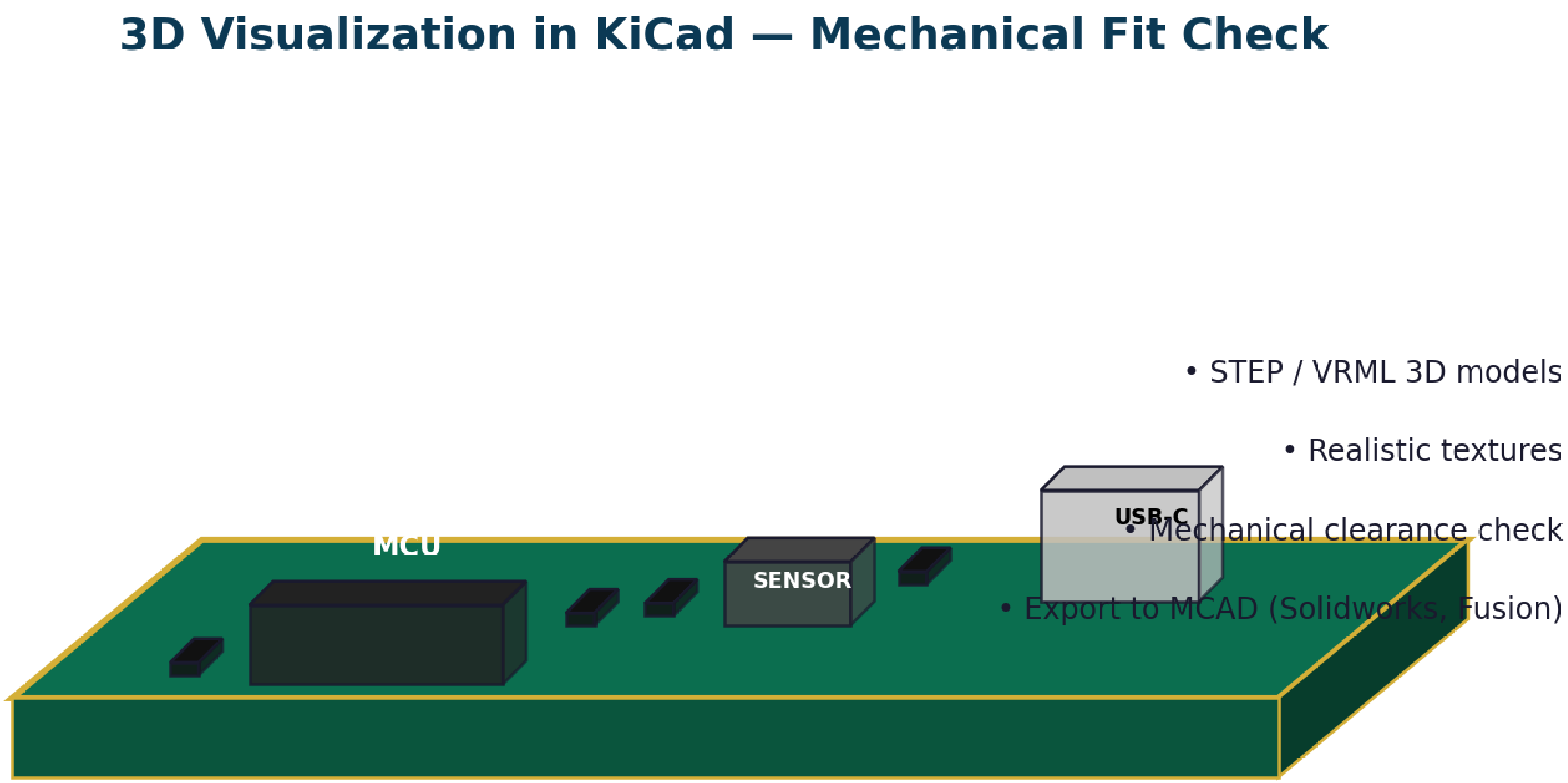
Gerber & Manufacturing Output Files

- **.GTL** — Top Copper
- **.GBL** — Bottom Copper
- **.GTS** — Top Soldermask
- **.GBS** — Bottom Soldermask
- **.GTO** — Top Silkscreen
- **.GBO** — Bottom Silkscreen
- **.GKO** — Board Edge
- **.DRL** — Drill File

Plus a Pick-and-Place (.pos) and BOM (.csv) for SMT assembly

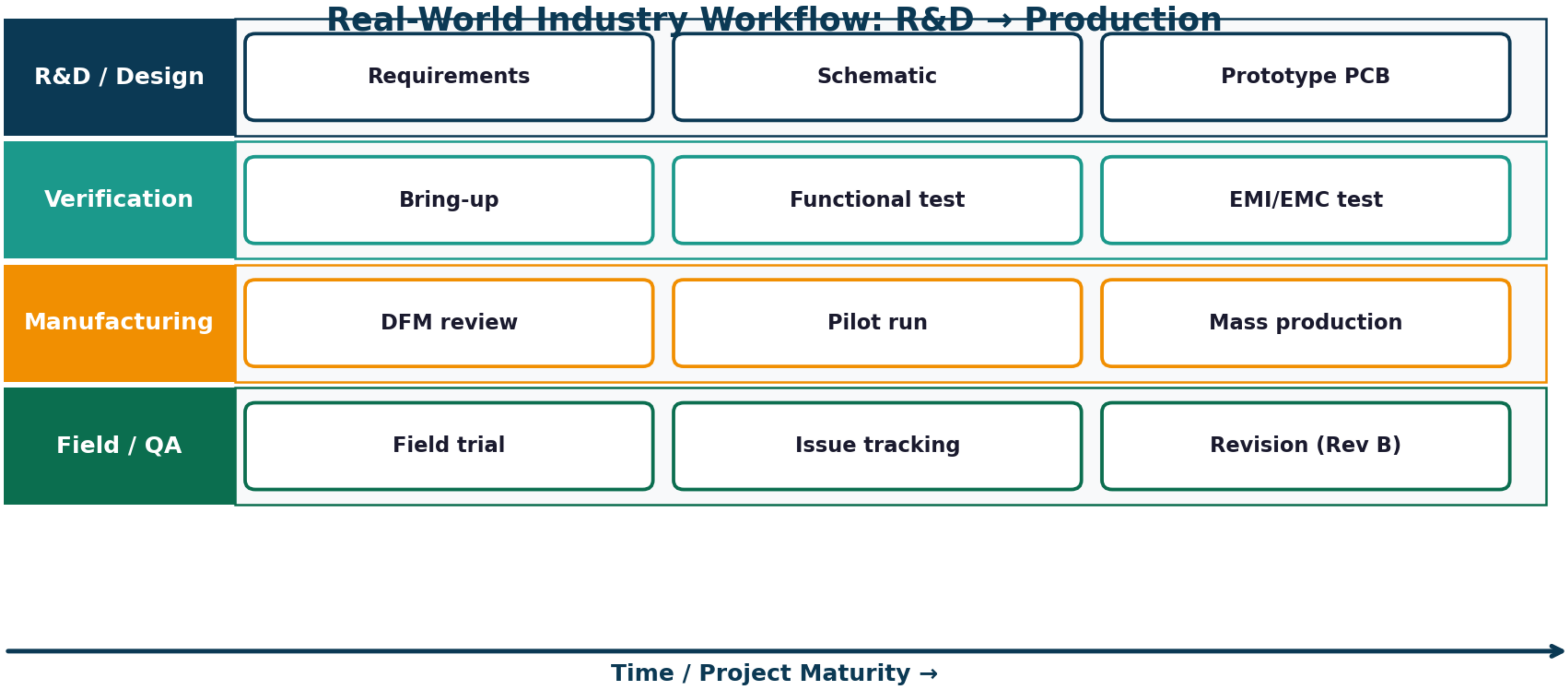
3D PCB Visualization in KiCad

Mechanical fit checks before fabrication



Real-world Industry Workflow

From R&D bench to mass production



Common Mistakes in PCB Design

Learn from the failures so you don't repeat them

Common Mistakes — and Why They Hurt

⚠ **Missing decoupling caps**	*Voltage spikes, MCU resets*
⚠ **90° trace corners**	*Acid traps in fab + impedance bump*
⚠ **Split ground planes**	*EMI, slow returns, noisy signals*
⚠ **No fiducials/test points**	*Hard to assemble & debug*
⚠ **Tiny silkscreen**	*Unreadable after fab*
⚠ **Wrong footprint**	*Board fails at assembly*
⚠ **Skipping DRC**	*Shorts, opens, scrap*
⚠ **Power traces too thin**	*Voltage drop, heating*

Best Practices

The habits that separate hobbyists from engineers

PCB Design Best Practices

✓ Start with a clean, hierarchical schematic	✓ Use a continuous ground plane on inner layer
✓ Place decoupling caps closest to power pins	✓ Match impedance for high-speed signals
✓ Use 45° (or curved) traces, never 90°	✓ Length-match clock & differential pairs
✓ Add test points & fiducial marks	✓ Keep analog & digital grounds separated, joined at one point
✓ Verify footprints against component datasheet	✓ Run ERC, DRC, and a manual review every revision
✓ Document every revision (Rev A → Rev B with notes)	✓ Build a 3D model for mechanical fit early

Future Trends in PCB Design

Where the industry is heading

Future Trends in PCB Design

1 HDI / Microvia

≤ 100 µm features
for mobile & wearables

2 Flex / Rigid-Flex

Foldable, IoT, medical implants

3 Embedded Components

R/C inside layers — denser boards

4 AI-Assisted Routing

ML-driven autorouters & placement

5 IPC-2581 / ODB++

Open formats replacing Gerber

6 Sustainable PCBs

Halogen-free, recyclable substrates

Conclusion

Key takeaways from today's session

Master the workflow

Schematic → ERC → Layout → DRC → Gerber
Iterate, don't ship Rev A.

KiCad is industry-ready

Free, open-source, Python-scriptable, used by startups and Fortune-500 alike.

Placement before routing

60% placement, 40% routing.
Good placement = signal integrity for free.

Ground plane = EMI insurance

Continuous reference plane on a 4-layer board solves most EMC headaches.

Footprints = source of truth

Verify every footprint against the datasheet.
It's the #1 cause of re-spins.

Practice on real boards

Design, fab, bring up, debug.
Three boards in your portfolio beats any certificate.

Questions & Discussion

Ask anything — design choices, KiCad workflow, industry practices, project ideas.

References & Further Learning

- kicad.org — official downloads, docs, and community forum
- docs.kicad.org — getting-started, tutorials, scripting reference
- snapeda.com • componentsearchengine.com — vetted symbols & footprints
- Phil's Lab (YouTube) — high-quality KiCad and signal-integrity walkthroughs
- Robert Feranec (YouTube) — advanced PCB & SI courses
- Henry Ott — Electromagnetic Compatibility Engineering (definitive EMC reference)
- Howard Johnson — High-Speed Digital Design (signal integrity classic)
- IPC-2221 / IPC-7351 — industry standards for PCB & footprint design

Thank you • Connect on LinkedIn / GitHub for follow-up questions